

Eliciting Willingness to Pay from Alternative Contingent Valuation Formats: An Application to a Large-Scale Water Diversion Project

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Abstract

This paper examines whether alternative contingent valuation method (CVM) formats elicit similar underlying willingness to pay (WTP) for non-market goods. We implement a survey experiment on the Han-to-Wei River Water Diversion Project, a major water infrastructure investment in China, by randomly assigning 1,218 respondents to double-bounded dichotomous choice (DBDC) or multiple-bounded discrete choice (MBDC) questionnaires. Using several estimation models, we estimate household WTP for improvements in water supply reliability and water quality. We find WTP is higher among households with greater income, stronger perceived water scarcity, poorer perceived water quality, and greater knowledge of local water systems. The estimated mean WTP is approximately 22 CNY per household per week, equivalent to 0.8% of monthly household income. After harmonizing bid support, the implied WTP distributions converge closely and center around the MBDC Not sure category, providing evidence for the reliability of CVM-based valuation of non-market goods.

Keywords: Contingent valuation; Willingness to pay; Water infrastructure projects; Experiment design.

JEL Codes: C93, Q51, Q25, C25

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1 Introduction

Contingent valuation methods (CVM) are widely used to elicit willingness to pay (WTP) for non-market goods when market prices are unavailable (Carson and Groves, 2007; Czajkowski et al., 2024). Conventional contingent valuation (CV) models typically assume that each respondent has a single latent WTP, accepting a bid below this value and rejecting one above it (Carson, 2012). However, respondents may face uncertainty in forming their own valuations, particularly for unfamiliar environmental goods and public projects, implying that individual WTP may instead be characterized by a valuation range or an individual-specific distribution (Shaikh et al., 2007; Akter et al., 2008; Voltaire et al., 2013).

A series of studies develops this view of valuation uncertainty. Wang (1997) argues that respondents are likely to report uncertainty when an offered bid lies near the boundary of their underlying WTP. Welsh and Poe (1998) develop the multiple-bounded discrete choice (MBDC) format, which allows respondents to evaluate a sequence of bids using ordered uncertainty responses. Subsequent studies formalize this framework by treating each individual’s WTP as a random variable characterized by individual-specific parameters (Wang and He, 2011; Wang et al., 2013; Su et al., 2024). Recent evidence from the response-time literature also shows that choice difficulty is associated with uncertainty and contains information about latent WTP (Alós-Ferrer et al., 2021; Liu and Netzer, 2023; Benkert et al., 2026). In the MBDC setting, this suggests that the transition around the Not sure category provides a benchmark for locating underlying WTP.

This perspective is important for assessing whether alternative CV formats elicit underlying valuations. Welsh and Poe (1998) show that WTP estimates from single-bounded dichotomous choice (SBDC) are broadly consistent with the valuation region implied by MBDC Not sure responses. SBDC is incentive compatible under appropriate conditions (Carson and Groves, 2007; Hanemann et al., 2025), whereas MBDC allows respondents to express valuation uncertainty directly. Their consistency supports the robustness of alternative CV formats to elicit the underlying WTP.

In this study, we extend the literature by comparing double-bounded dichotomous choice (DBDC) with uncertainty-inclusive MBDC responses. DBDC is widely used in applied CV research for its statistical efficiency gains from the follow-up bid (Hanemann et al., 1991; Kanninen, 1993; Carson and Groves, 2007; Carson, 2012), although the introduction of the second bid may

also alter respondents' information set and subsequent responses (Carson and Groves, 2007). This comparison provides evidence on the robustness of DBDC in eliciting underlying WTP and, together with the existing SBDC–MBDC evidence, allows a more comprehensive assessment of the reliability of WTP estimates across the three commonly used CV frameworks.

To this end, we conduct a survey experiment that randomly assigns respondents to DBDC or MBDC questionnaires. Using the Han-to-Wei River Water Diversion Project in Shaanxi, China, we examine how residents value a major inter-basin water infrastructure project and whether alternative CV formats generate similar valuation estimates. The Han-to-Wei River Water Diversion Project is a major inter-basin water transfer initiative designed to alleviate chronic water scarcity in the Guanzhong Plain. While the project is expected to improve water supply reliability and water quality, it also entails substantial fiscal, environmental, and social costs (Zhao et al., 2025). This setting provides a useful application for estimating household WTP for a large-scale public infrastructure project.

We model respondents' WTP as a function of demographic, economic, and water-related characteristics and estimate the specification using three approaches: a DBDC model, an MBDC model proposed by Welsh and Poe (1998) (hereafter Welsh-type model), and an MBDC model following Wang and He (2011), Wang et al. (2013), Wang and He (2018), and Su et al. (2024) (hereafter Wang-type model). Across the three models, the determinants of WTP are broadly consistent. We find that WTP is generally lower among women and older respondents, higher among higher-income households, and strongly associated with perceived water scarcity, water quality, and knowledge of local water conditions.

Beyond the determinants of valuation, we compare the implied WTP distributions across elicitation formats. The DBDC questionnaire includes bids up to 30 CNY, whereas the MBDC survey employs two bid designs in separate samples. The main MBDC design extends the bid range to 300 CNY, allowing respondents to express valuations over a wider support and reducing the risk of truncating the upper tail of the WTP distribution (Welsh and Poe, 1998; Alberini et al., 2003; Vossler et al., 2004). To facilitate a direct comparison with DBDC, we also implement a capped MBDC design with the same 30 CNY upper bound. Under the broader MBDC range, the implied WTP distribution exhibits a heavier upper tail than the DBDC estimates, reflecting the additional valuation range available in the MBDC questionnaire. Once the bid support is harmonized,

however, the estimated distributions converge closely, with the DBDC estimates aligning with the valuation region implied by the MBDC Not sure responses. This convergence indicates that the apparent cross-format differences are largely attributable to differences in bid support and provides evidence that DBDC can robustly elicit respondents' underlying WTP when the survey designs are made comparable.

Our study makes three contributions to the literature. First, the paper provides one of the few integrated evaluations of CV formats within the same survey environment (Alberini et al., 2003; Voltaire et al., 2019; Ahi et al., 2023; Czajkowski et al., 2024; Skeie et al., 2025; Thapa et al., 2025), and contributes more broadly to comparisons between valuation strategies (He et al., 2017; Sandorf and Navrud, 2026). The convergence between DBDC and MBDC Not sure estimates directly extends the consistency documented by Welsh and Poe (1998) between SBDC and uncertainty-inclusive MBDC responses to the double-bounded setting, despite concerns that follow-up bids may affect subsequent choices (Carson and Groves, 2007). More broadly, if allowing respondents to express uncertainty yields WTP estimates that are closer to their true valuations, the convergence of estimates from alternative models toward the WTP implied by the MBDC Not sure category indicates that models produce broadly consistent valuations and elicit respondents' underlying WTP. This finding alleviates the concern raised by Bond and Lang (2019) that welfare estimates derived from discrete survey responses may be driven primarily by researchers' identifying assumptions instead of respondents' actual preferences, strengthening the case for the reliability of CVM in valuing non-market goods.

Second, the paper contributes to the debate on whether alternative CV formats generate systematically different preference estimates, including bid range selection (Cook et al., 2018; Glenk et al., 2024; Börger et al., 2025), hypothetical bias (Loomis, 2011; Mjelde et al., 2012; Liu and Tian, 2021; Ahles et al., 2024; Chopra and Vossler, 2026), and starting-point bias (Herriges and Shogren, 1996; Carson, 2012; López-Mosquera and Sánchez, 2012; Carson et al., 2014). By comparing the full WTP distributions, we show that the apparent differences between models are largely driven by discrepancies in bid support. These findings suggest that commonly used CV models, despite their parametric and distributional assumptions, can generate coherent and comparable welfare measures when survey design choices are carefully aligned (Bond and Lang, 2019; Alós-Ferrer et al., 2021; Liu and Netzer, 2023). Thus, concerns over model specification and response behavior do not

invalidate stated preference methods but instead highlight the importance of design discipline and cross-format validation.

Third, the paper provides evidence on the determinants of WTP for large-scale water diversion projects. The credible welfare valuation of such projects remains relatively rare in the literature, yet it carries important policy relevance and welfare consequences given their scale, complexity, and long-term externalities (Wang et al., 2013; Du et al., 2023). This paper helps fill this gap by incorporating both economic and social factors into the analysis of WTP, providing policy-relevant welfare estimates that are directly informative for cost–benefit analysis and infrastructure design.

The remainder of the paper is organized as follows. Section 2 describes the background of the Han-to-Wei Water Diversion Project. Section 3 introduces the survey design based on DBDC and MBDC, the estimation strategy, and the descriptive statistics. Section 4 presents the results of the estimation of WTP determinants and the comparison of the models. Section 5 reports robustness checks and additional discussion. Section 6 concludes.

2 Background of Han-to-Wei River Water Diversion Project

The Han-to-Wei River Water Diversion Project is one of the most important inter-basin water transfer projects in western China. Located in Shaanxi Province, the project diverts water from the Han River, a major tributary of the Yangtze River in southern Shaanxi, to the Wei River Basin, which belongs to the Yellow River system in the north. Initially proposed in the 1950s, the project was officially launched in 2011 as part of China’s broader effort to address the chronic north–south water imbalance.

The primary beneficiary of the diversion is the Guanzhong Plain, the economic and political center of northwest China. The region has long faced severe water scarcity due to low precipitation, limited groundwater recharge, and growing demand from agriculture, industry, and urbanization. With a total investment of more than 51.6 billion CNY (7.17 billion USD), the project is designed to transfer approximately 1.5 billion cubic meters of water annually from the Han River Basin to the Wei River Basin, benefiting more than 14 million residents across roughly 14,000 square kilometers. The project spans more than 300 kilometers and includes reservoirs, tunnels, pumping stations, and canals. Its most technically challenging component is the crossing of the Qinling

Mountains, including the 98.3-kilometer Qinling water conveyance tunnel. Figure 1 illustrates the geographical layout of the project.

The project is expected to generate far-reaching benefits. At the macro level, it supports regional economic integration, industrial upgrading, and balanced development in Shaanxi Province. By stabilizing the water supply and improving the ecological environment, it improves water security and climate resilience in the semi-arid regions of northwestern China (Liu and Yang, 2012). At the micro level, the project substantially improves household and agricultural water availability, reduces over-extraction of groundwater, and lowers irrigation costs. It directly benefits millions of residents through more reliable domestic water use, improved water quality, and better living and sanitation conditions, generating long-term environmental and social benefits (Zhao et al., 2015).

Nevertheless, the project may also impose economic and social costs on local communities. For residents, the most direct impact arises from increased fiscal burdens introduced to cover part of operating and maintenance costs. In addition, the diversion process can also alter local ecosystems, creating potential risks to riverine biodiversity and agricultural water balance in both the source and the receiving areas (Knapp et al., 2003). From a social perspective, construction activities have caused temporary disruptions in transport and local livelihoods, while long-term benefits may not be evenly distributed between regions and income groups (Sun et al., 2021). These multidimensional costs shape public perceptions and determine the degree of social acceptance of the project.

Understanding how local residents evaluate perceived trade-offs between improvements in water availability and reliability and the associated economic, environmental, and social costs is essential. In this context, this study uses the CVM to estimate the public’s WTP for the Han-to-Wei River Water Diversion Project. By quantifying residents’ economic and psychological valuations of tangible benefits of water-use versus the costs they perceive, this research provides empirical support for cost–benefit analysis, policy design, and sustainable management of water resources.

3 Survey Design and Estimation Methods

In this section, we first describe the survey design based on the DBDC and MBDC frameworks, together with the corresponding maximum likelihood estimation (MLE) approach. We then present the survey implementation and provide descriptive statistics of the sample.

3.1 DBDC

The DBDC extends the single-bounded model by presenting each respondent with two sequential valuation questions. In the first question, respondents are asked whether they would be willing to pay an initial offer. Conditional on the initial response, a follow-up question is then posed with either a higher or a lower bid. This two-stage design narrows the interval that contains the true WTP of the respondent, improves statistical efficiency, and yields more precise parameter estimates compared to the single-bounded approach (Carson and Groves, 2007). An example of the two valuation questions used in our survey is provided below.

Question 1 *To address the problems of substandard water quality for residents and insufficient industrial water supply across the province, it is necessary to implement a water diversion project that transfers water from the Yangtze River. If such a project is successfully completed, the quality of drinking water sources will reach Class II standards, which is significantly higher than the current level, and both residential water supplies will be reliably provided without major interruptions in the future.*

*We would like to understand your evaluation of such a project. Considering your household's income, expenditures, and financial capacity, **would your family be willing to pay 2 CNY each week** to support the construction and operation of this project?*

A. Yes

B. No

If the respondent chooses **Yes** in Question 1, a second question with a higher bid amount is presented:

Question 2 *Furthermore, **would your family be willing to pay 4 CNY each week** to support the construction and operation of this project?*

A. Yes

B. No

Alternatively, if the respondent chooses **No** in Question 1, we will decrease the bid amount:

Question 3 *Furthermore, would your family be willing to pay 1 CNY each week to support the construction and operation of this project?*

A. Yes

B. No

To mitigate starting-point bias and improve identification efficiency, we employ four different bid combinations in the DBDC questions and randomly assign each combination to respondents. The four bid combinations are shown in Table 1, with the last column reporting the abbreviation for each combination. The highest bid amount in DBDC is 30 CNY.

This DBDC design allows us to identify the upper and lower bounds of the individual WTP. Based on these bounds, we construct the likelihood function and estimate the parameters of the WTP distribution under specific parametric assumptions. Following the literature with DBDC applications (Rodemeier, 2026), first, we use the following standard functional form to capture the WTP of each respondent i in terms of individual or household characteristics:

$$\ln WTP_i = x_i' \beta + \varepsilon_i$$

where x_i is a vector of characteristics. ε_i is an error term that is assumed to follow a logistic distribution with mean zero and standard deviation σ .

The log form of WTP is important for two reasons. First, CVM analyses generally rely on predicted WTP values, which must be non-negative. Second, taking logs helps address potential skewness in the WTP distribution. For these reasons, we use the log specification in all models throughout the paper.

Next, because each respondent answers two valuation questions, we categorize the response options as YY, YN, NY, and NN, where the first letter denotes the response to the initial bid, and the second letter denotes the response to the follow-up bid, with Y for Yes and N for No. The follow-up question is contingent on the initial response of the respondent.

For example, in the bid combination 2–4–1, YY (YN) indicates that the respondent accepts 2 CNY and accepts (rejects) 4 CNY, while NY (NN) indicates that the respondent rejects 2 CNY but accepts (rejects) 1 CNY. Given the resulting WTP bounds implied by these response options and assuming individuals' WTP is drawn from a population distribution of WTP, we write the

log-likelihood function as follows:

$$l(\beta, \sigma) = \sum_{i=1}^n \left[I_{YY,i} \ln P_{YY,i} + I_{YN,i} \ln P_{YN,i} + I_{NY,i} \ln P_{NY,i} + I_{NN,i} \ln P_{NN,i} \right] \quad (1)$$

where $I_{AA,i}$ is an indicator function equal to one if respondent i exhibits a response AA ($A \in \{Y, N\}$), and zero otherwise. $P_{AA,i}$ is the probability of choosing the response AA given the parameters. By assuming log of WTP follows the logistic distribution, we write:

$$\begin{aligned} P_{YY,i} &= 1 - \Lambda \left(\frac{\ln b_{h,i} - x'_i \beta}{\sigma} \right) \\ P_{YN,i} &= \Lambda \left(\frac{\ln b_{h,i} - x'_i \beta}{\sigma} \right) - \Lambda \left(\frac{\ln b_{m,i} - x'_i \beta}{\sigma} \right) \\ P_{NY,i} &= \Lambda \left(\frac{\ln b_{m,i} - x'_i \beta}{\sigma} \right) - \Lambda \left(\frac{\ln b_{l,i} - x'_i \beta}{\sigma} \right) \\ P_{NN,i} &= \Lambda \left(\frac{\ln b_{l,i} - x'_i \beta}{\sigma} \right) \end{aligned}$$

where $b_{h,i}$ and $b_{l,i}$ are the higher and lower follow-up bids, respectively. $b_{m,i}$ is the first bid. Given the estimated parameters, we can estimate the implied WTP distribution.

3.2 MBDC

The MBDC extends the double-bounded approach by presenting respondents with a sequence of bid amounts. By introducing additional bid levels into the questionnaire, MBDC can improve the statistical efficiency of the estimates, as noted by [Welsh and Poe \(1998\)](#). For each bid, respondents indicate whether they are willing to pay the stated amount for the good or policy proposed. This design yields multiple valuation cutoffs for each individual, allowing the respondent's true WTP to be more tightly bracketed. As a result, the MBDC format improves statistical efficiency and reduces information loss relative to the single- and double-bounded models.

Table 2 summarizes the survey design, with bids ranging from 0 to 300 CNY. We adopt a higher upper bound than the 30 CNY used in the DBDC format to better capture the upper tail of the WTP distribution ([Welsh and Poe, 1998](#); [Alberini et al., 2003](#); [Vossler et al., 2004](#)). The implementation of a wide bid range is more constrained in the DBDC format because the follow-up

bid is determined by the response to the initial bid. Extending the overall bid support may therefore require substantially higher or lower bid levels at different stages of the elicitation sequence, potentially increasing susceptibility to anchoring, starting-point effects, and other response distortions. Accordingly, the DBDC design employs a narrower bid range. To distinguish differences attributable to elicitation format from those arising from bid support, we additionally include a smaller MBDC sample with bids capped at 30 CNY, as discussed in Section 3.4.

Our questionnaire follows [Welsh and Poe \(1998\)](#) and allows respondents to express five ordered uncertainty categories for each stated bid: Definitely yes, Probably yes, Not sure, Probably no, and Definitely no. In particular, the Not sure category is designed to capture the greatest degree of valuation uncertainty, since it does not require respondents to lean toward either acceptance or rejection of the stated bid.

Welsh-type Model. For estimation, we compare two models proposed by [Welsh and Poe \(1998\)](#) and [Wang and He \(2011\)](#). The Welsh-type model has been widely used in the MBDC valuation ([Carlsson et al., 2022](#); [Beasley and Dundas, 2026](#)). The model incorporates response uncertainty by treating each certainty category as defining a distinct threshold of the latent WTP distribution. For each category w , the sequence of responses across bid levels identifies an interval within which the corresponding WTP threshold lies. Specifically, the highest bid associated with a response at or above a given certainty level and the lowest bid associated with a response below that level determine the lower and upper bounds, $b_{l,i,w}$ and $b_{u,i,w}$. The five response categories therefore generate separate interval-valued measures of WTP associated with different degrees of certainty.

According to [Welsh and Poe \(1998\)](#), the log-likelihood function is as follows given the knowledge of the upper bound $b_{u,i,w}$ and the lower bound $b_{l,i,w}$ for each of the five uncertainty categories w :

$$l^w(\beta_w, \sigma_w) = \sum_{i=1}^n \ln \left[\Lambda \left(\frac{\ln b_{u,i,w} - x'_i \beta_w}{\sigma_w} \right) - \Lambda \left(\frac{\ln b_{l,i,w} - x'_i \beta_w}{\sigma_w} \right) \right] \quad (2)$$

where $\Lambda(\cdot)$ is the cumulative logistic distribution function.

Wang-type Model. An alternative approach is the Wang-type model proposed by [Wang and He \(2011\)](#) which is adopted in the recent literature on valuation ([Wang et al., 2013](#); [Bigerna and Polinori, 2014](#); [Su et al., 2024](#)). This model follows a two-step estimation strategy. In the first

step, individuals' WTP is modeled as a random variable with an individual-specific distribution, and the parameters of this distribution are estimated using MBDC responses. In the second step, the estimated mean WTP is used to examine how socioeconomic characteristics and perceptions explain the WTP between individuals.

Following Wang and He (2011), in the first step, we assume that

$$\ln WTP_i \sim \text{logistic}(\mu_i, \sigma_i)$$

where μ_i and σ_i are the location and scale parameters of the individual-specific WTP distribution.

For each bid amount b_{ij} , the respondent i reports an ordered uncertainty category that is assigned to a numerical probability $P_{ij} \in (0, 1)$. This probability is interpreted as the respondent's subjective assessment of

$$P_{ij} \equiv P(WTP_i > b_{ij}) = 1 - \Lambda\left(\frac{\ln b_{ij} - \mu_i}{\sigma_i}\right) + \lambda_i$$

where in the baseline model, we follow Wang and He (2011) and assign probabilities of 0.999, 0.75, 0.5, 0.25, and 0.001 to the five categories, ranging from Definitely willing to Definitely unwilling, respectively, in Table 2. The reporting error satisfies $\lambda_i \sim \text{logistic}(0, \delta)$.

Therefore, the log-likelihood function is:

$$l_i(\mu_i, \sigma_i, \delta) = \sum_j \ln \left[f_\lambda \left(\frac{P_{ij} - 1 + \Lambda\left(\frac{\ln b_{ij} - \mu_i}{\sigma_i}\right)}{\delta} \right) \right] \quad (3)$$

Maximizing this likelihood yields estimates of the individual-specific WTP distribution parameters (μ_i, σ_i) . These parameters characterize the WTP distribution for each respondent.

In the second step, the estimated individual-level WTP moments are used as dependent variables in regression models to examine how socioeconomic characteristics and perceptions explain WTP:

$$\hat{\mu}_i = x_i' \beta + e_i \quad (4)$$

To summarize, the standard DBDC model assumes that each respondent has a single latent WTP point, with the observed yes/no responses providing interval information about the location

of that point; the estimated distribution reflects heterogeneity in latent WTP *across respondents*. In contrast, the MBDC approaches allow uncertainty to enter at the *individual level*. The Welsh-type model uses ordered certainty responses to construct category-specific intervals for latent WTP thresholds, while the Wang-type model explicitly treats each respondent’s WTP as a random variable characterized by an individual-specific location and scale parameter.

3.3 Consistency of DBDC and MBDC

Given the estimates obtained from the DBDC and MBDC specifications, we examine whether different CV formats yield statistically significant differences in estimated WTP and its determinants. We consider two approaches. First, we use Kolmogorov–Smirnov (KS) test to compare WTP distributions in different models (Gulati et al., 2024; Roth et al., 2024). Specifically, we construct the following statistics:

$$KS = \sqrt{\frac{n_1 n_2}{n_1 + n_2}} \cdot \sup_{WTP} |F_1(WTP) - F_2(WTP)| \sim \text{Kolmogorov distribution} \quad (5)$$

where subscripts 1 and 2 respectively denote any two estimation models among DBDC, Welsh-type, and Wang-type. n_1 and n_2 are the corresponding sample sizes, respectively. $F(WTP)$ represents the empirical cumulative distribution function of the predicted WTP implied by each model. If the KS statistic is not statistically significant, we cannot reject the null hypothesis that the two implied WTP distributions are equal.

A challenge in our setting is that the WTP distributions are not directly observed but are generated from estimated models. As a result, the asymptotic distribution of the KS statistic may deviate from the standard Kolmogorov distribution. To address this issue, we implement a bootstrap procedure to obtain a valid inference for the KS test.

Second, we employ a Wald test to examine whether the model parameters are statistically different in any two estimation models (Brouwer et al., 2017). The Wald statistic is constructed as follows:

$$W = (\hat{\beta}_1 - \hat{\beta}_2)' Var(\hat{\beta}_1 - \hat{\beta}_2)^{-1} (\hat{\beta}_1 - \hat{\beta}_2) \sim \chi^2(r) \quad (6)$$

where subscripts 1 and 2 denote two estimation models, respectively. $\hat{\beta}_1$ and $\hat{\beta}_2$ are estimates of

parameters for each model. If the Wald statistic is not statistically significant, we cannot reject the null hypothesis that the parameter estimates are the same across models.

The main challenge of this approach is that a plug-in estimator for $Var(\hat{\beta}_1 - \hat{\beta}_2)$ is not available. To address this issue, we implement a bootstrap procedure to obtain a consistent estimate of the variance¹.

3.4 Survey Implementation

The survey was conducted in Xi'an City, the largest and most economically developed city that relies on the Wei River as a source of public water supply. In-person interviews were conducted in 2023 and 2024 by trained graduate students in economics from Xi'an Jiaotong University. The respondents were selected using a stratified sampling strategy in the city's six largest districts, with strata defined to match the distribution of the population in terms of age, gender, and educational level.

The questionnaire consists of three parts. In the first part, respondents were asked to sign the following commitment statement before answering the survey questions.

Commitment 1 *I pledge that I will carefully consider and truthfully evaluate the following questions in this survey. I will provide information to the best of my knowledge and answer all questions seriously and honestly.*

The oath script can help reduce the hypothetical bias in CVM (Vossler et al., 2025). In the main analysis, we use the full sample of respondents who signed the commitment statement.

The second part of the survey involved warm-up questions and collected information on respondents' individual and household characteristics, including age, gender, educational level, household size, and monthly household income. It also included questions about water-related conditions, such as perceived water quality, water prices, and concerns about water pollution.

In the final part of the survey, respondents' WTP was elicited using the DBDC or MBDC format, following an introduction to the background of the survey that explicitly described the current shortage of water supply in Xi'an City, as shown in the Supplementary Materials. The

¹Another question of interest is the relative statistical efficiency of the models. In Appendix G, we introduce the comparison of statistical efficiency across the three models and provide some discussions.

respondents were randomly assigned to one of the two elicitation methods and, aside from the CVM elicitation format, the two groups received identical questionnaires. After quality control procedures, the final sample used for the main analysis consists of 610 respondents in the DBDC group and 422 respondents in the MBDC group. The complete questionnaire of our survey is included in the Supplementary Materials.

As noted above, the primary MBDC design uses bids up to 300 CNY to capture a wider range of valuations. This main MBDC sample contains 422 respondents and is used for the baseline analysis. Because its bid range substantially exceeds the 30 CNY upper bound in the DBDC design, we also surveyed a separate capped-bid MBDC group with bids restricted to 30 CNY. This group contains 97 respondents and is used specifically to harmonize bid support when comparing WTP distributions across elicitation formats.

We surveyed an additional MBDC group using the full 300 CNY bid range but without the oath script. This group contains 89 respondents and is used only to examine the role of the oath in mitigating hypothetical bias. It is excluded from the baseline analysis and discussed separately in Section 5.

3.5 Descriptive Statistics

Descriptive statistics are reported in Table 3 with detailed categorical statistics in Appendix Tables A1 to A3. We find most variables are balanced between groups, indicating that the random assignment was successful. We also report several corresponding statistics from the city census to assess the representativeness of our sample. In general, our sample closely matches the census distribution, with no substantial differences observed.

The descriptive statistics also characterize respondents' water-related conditions and perceptions. With respect to water scarcity, we do not observe severe constraints in access to tap water within the city. In particular, the mean frequency of water outages is less than one occurrence and the average duration of the outage is less than one hour. In contrast, respondents report relatively low satisfaction with water quality. The mean scores for perceived tap water impurity and overall water quality are below the midpoint of the questionnaire scale. Finally, in terms of knowledge of water sources, respondents appear to be more responsive to cognitive aspects of water-related information, such as awareness of water pricing and concerns about water pollution.

It is important to note that within the categories of water scarcity, water quality, and knowledge of water sources, several variables capture closely related aspects. To address potential multicollinearity and reduce dimensionality, we apply principal component analysis (PCA) and construct a single index for each category. Details of the PCA results are reported in the Appendix Tables [B1](#) to [B3](#). In summary, we construct three new indices (water scarcity index, water quality index, and water knowledge index). A higher value indicates a higher level of the corresponding attribute.

To conclude the introduction of our survey, we present the distribution of responses to the DBDC and MBDC bid questions in Appendix Tables [A4](#) and [A5](#). As expected, lower bid amounts are associated with a higher concentration of yes responses in the DBDC format and a greater mass of willing responses in the MBDC format. These results are consistent with economic intuition and indicate that bid design and elicitation formats function as intended.

4 Results and Discussion

This section presents the estimation results in two parts. First, we report the results of the DBDC and MBDC specifications to examine the determinants of WTP for the water diversion project. Second, we describe the implied distribution of the WTP and compare the estimates of the three models.

4.1 Determinants of WTP

Table [4](#) reports the DBDC estimates of the WTP function based on the likelihood specified in equation [\(1\)](#). Since the dependent variable is specified in logarithmic form, the coefficients can be interpreted approximately as percentage changes in WTP.

Several characteristics of the household and the respondents are associated with WTP. The age coefficient is negative and marginally significant, with an estimated coefficient of -0.007, implying that each additional year of age is associated with approximately a 0.7 percent lower WTP. This finding suggests that older respondents tend to place less value on the long-term benefits of the diversion project, possibly reflecting stronger discounting of future gains or greater aversion to uncertainty ([Chan et al., 2019](#)). The coefficient on female is -0.349 and statistically significant, indicating that female respondents report a WTP that is approximately 35 percent lower than that

of male respondents, keeping other factors constant. This gender gap may reflect differences in intra-household decision-making roles, risk attitudes, or preferences toward long-term infrastructure projects with delayed benefits (Dittrich and Leipold, 2014; Cohen et al., 2020; Su et al., 2024).

In contrast, having a college degree does not appear to have a statistically significant effect on WTP, with the estimated coefficient being positive but statistically indistinguishable from zero. Household income enters positively and is highly statistically significant, with an estimated coefficient of 0.014, suggesting a positive income effect and indicating that improved water security is a normal good. Monthly water expenditure is also positive and statistically significant, with an estimated coefficient of 0.011, implying that households with higher baseline spending on water tend to place a higher value on the diversion project.

Water-related variables play an important role in the valuation. Households who have installed water filters at home exhibit a substantially higher WTP, with an estimated coefficient of 0.791, consistent with stronger concerns about water quality. The water scarcity index enters positively and is highly significant, indicating a higher value among households facing more frequent or more severe supply disruptions. The water quality index is negative and significant, with an estimated coefficient of -0.271; given the construction of the index, this implies that households perceiving poorer tap-water quality are willing to pay more for the project. Finally, the water knowledge index is positive and significant, with an estimated coefficient of 0.166, suggesting that greater awareness of water sources, pricing, and pollution issues is associated with a higher valuation of the diversion project. These results are consistent with existing studies on the valuation of improvements in water quality and related ecosystem services (Kayamo, 2022; Viti et al., 2023).

The results documented above have several implications for the design of large-scale water diversion projects. The positive income coefficient suggests that uniform cost-recovery mechanisms based on water tariffs can raise distributional concerns, pointing to the potential role of differentiated pricing or complementary fiscal arrangements (Abu Hatab et al., 2025; Moon and Choi, 2026). At the same time, a higher WTP among households facing greater water scarcity and poorer perceived water quality indicates that perceived exposure to water-related risks is strongly correlated with public support, implying that prioritizing service improvements in more water-stressed areas can improve both efficiency and social acceptance (Chatterjee et al., 2017; Guerrini et al., 2018). Finally, the positive association between water-related knowledge and WTP highlights the

importance of information and transparency in the valuation, suggesting that clearer communication about water sources, pricing, and environmental conditions may strengthen public support for long-term infrastructure investments (Kayamo, 2022).

Table 5 uses the MBDC sample with bids up to 300 CNY and presents the Welsh-type MBDC estimates in the five uncertainty categories. We do not find significant differences between the different uncertainty categories. The similarity across categories suggests that covariate effects are broadly stable across different uncertainty thresholds, with little evidence that their associations with WTP vary systematically across categories. Compared with DBDC results, demographic estimates remain consistent. Age and female enter negatively, while income enters positively. In contrast to the DBDC results, having a college degree now becomes more positively associated with WTP.

Although water expenditure and filter installation are not statistically significant, the three water-related indices remain statistically significant across the uncertainty categories, reinforcing the importance of both perceived water conditions and cognitive awareness in shaping stated preferences.

Finally, Appendix Table C1 and Table 6 report the Wang-type MBDC estimates. According to the first-stage results in Appendix Table C1, the mean WTP is 22 CNY, with a standard deviation of 20 CNY. The second-stage regression results in Table 6 are broadly consistent with the findings of the DBDC and Welsh-type specifications. Income, education, and water knowledge are positively associated with WTP, while age and female remain negative. The water scarcity, quality, and knowledge indices retain their expected signs and strong statistical significance.

Across the three estimation models, most demographic and water-related variables enter with similar signs and comparable levels of statistical significance across the DBDC and both MBDC specifications, showing that the main conclusions are robust to alternative elicitation formats.

4.2 Comparison of Estimates

In this subsection, we present the distributions of the estimated WTP (log) obtained from the three models in Figure 2. Panel (a) compares the log WTP distributions implied by the Welsh-type and Wang-type MBDC specifications and provides evidence of the close similarity between the two models. First, the distribution generated by the Wang-type model lies entirely within

the intermediate categories, close to the Not sure category, without exhibiting systematic shifts toward either extreme ones. Second, the two distributions display nearly identical shapes. In both cases, the survival function declines sharply as the log WTP increases over the 1 to 4 range and approaches zero near 5, with virtually indistinguishable tails.

Next, we compare the WTP distributions implied by the Welsh-type MBDC and the DBDC specifications. To facilitate a meaningful comparison, we proceed in two steps. We first plot the estimated survival functions using the original bid structures from the MBDC with bids up to 300 CNY and DBDC elicitation formats in Panel (b). In this comparison, the DBDC distribution exhibits a steeper decline and a lower overall range of WTP relative to the MBDC curves.

However, this difference should be interpreted with caution and should not be taken as direct evidence of systematic discrepancies between estimation models. The two elicitation formats differ substantially in their bid designs. In particular, the MBDC design includes bid values that extend to 100 to 300 CNY, while the DBDC module restricts bids to a maximum of 30 CNY. This asymmetry in bid ranges raises an important concern for comparability and introduces bias from bid support (Kanninen, 1995). The observed differences in distributional shapes may partly reflect differences in bid range selection rather than genuine differences in valuation.

To address this concern, we implement an adjustment in Panel (c). Specifically, we use the additional small MBDC group capped at 30 CNY and re-estimate the model parameters using maximum likelihood. This adjustment ensures that both elicitation formats are identified over a comparable support of bid levels, removing the influence of the extended upper tail in the original MBDC design.

After re-estimation, the resulting WTP distributions exhibit highly similar shapes. Moreover, the DBDC distribution is squarely in the Not sure category implied by the MBDC estimates. This result indicates that the apparent differences observed under the original bid structures are largely driven by discrepancies in bid design rather than fundamental differences in valuation. Once evaluated on a common bid support, the two elicitation models yield broadly consistent welfare estimates.

Finally, Panels (d) and (e) compare the WTP distributions implied by the Wang-type MBDC method and those estimated from the DBDC responses. Panel (d) presents the comparison based on the original bid structures. Under this specification, the survival function implied by the DBDC

format declines more dramatically and exhibits a narrower support relative to the distribution estimated from the Wang-type method. Panel (e) repeats the comparison after harmonizing the bid support by using the capped MBDC group. Once the bid ranges are aligned, the two estimated distributions become nearly identical in both shape and support.

This convergence is the key methodological finding. Among the five MBDC response categories, the Not sure category is intended to capture the greatest degree of valuation uncertainty. Unlike the Probably yes or Probably no responses, it does not require respondents to lean toward either acceptance or rejection of the stated bid. Recent studies suggest that allowing respondents to express this uncertainty may generate WTP estimates that are closer to their true valuations, because respondents are not forced to report an affirmative or negative preference when their valuation is ambiguous (Liu and Netzer, 2023; Benkert et al., 2026). We view the WTP implied by the MBDC Not sure category as a particularly informative benchmark for respondents' true valuations. After harmonizing bid support, the estimates from the DBDC model and both MBDC specifications converge around this benchmark.

This finding also connects our evidence to earlier comparisons between single-bounded dichotomous choice and MBDC. Welsh and Poe (1998) show that WTP estimates based on MBDC uncertainty responses are close to those obtained from single-bounded dichotomous-choice questions. Our results extend this evidence by showing that DBDC estimates and the two MBDC estimates also converge toward the WTP implied by the Not sure category once bid ranges are aligned. If uncertainty responses provide estimates closer to true WTP, this convergence suggests that alternative CVM formats can, to a substantial extent, elicit respondents' underlying valuations when survey designs are comparable.

At the same time, we conclude that the variation in the estimated WTP between formats can be traced primarily to differences in the bid design if the bid range is not harmonized (Cook et al., 2018; Glenk et al., 2024). When the support of bid values is standardized across DBDC and MBDC, the resulting WTP distributions become highly similar. This finding indicates that different elicitation formats largely differ in how stated preferences are recorded, rather than in the preferences they uncover. More generally, our findings imply that widely used CVM models can generate consistent and comparable welfare estimates when survey design choices are carefully coordinated, even though these models rely on parametric and distributional assumptions that have

been subject to criticism (Alós-Ferrer et al., 2021; Liu and Netzer, 2023).

The visual evidence above suggests the consistency of different models. Moreover, we follow equations (5) and (6) to perform pairwise KS and Wald tests between the three models in Tables 7 and 8, respectively. We cannot reject the null hypothesis that the estimates from the Wang-type MBDC, the Welsh-type MBDC in the Not sure category, and the DBDC model are equal, especially after adjusting the bid range.

4.3 Comparative Discussion and Cost-Benefit Analysis

Table 9 summarizes existing valuation studies on water infrastructure and related water-service improvements. Overall, our estimates are broadly comparable to the literature. Most studies report WTP values below 1% of monthly household income. In our baseline Wang-type MBDC estimate, the implied WTP corresponds to about 0.8% of average monthly household income in our sample². This relatively high estimate is reasonable given the scale and scope of the Han-to-Wei River Water Diversion Project. Unlike studies focusing on narrower improvements, such as tap-water quality or leakage reduction, this project is a large-scale inter-basin water diversion project intended to improve both water supply reliability and water quality. The implied income share is therefore near the upper end of existing estimates, but remains comparable to recent evidence on water security and nature-based infrastructure.

To conclude the main analysis, we conduct a cost-benefit comparison for the Han-to-Wei River Water Diversion Project. This analysis provides a useful benchmark for interpreting the estimated WTP as the project involves substantial upfront construction investment, while its benefits accrue gradually through improved water security, better water quality, and more reliable water supply.

We draw on the framework of Finkelstein and Hendren (2020) to analyze the net benefit. Let $\overline{WTP}_{week}$ denote the mean weekly WTP per household, and let N denote the number of beneficiary households. We approximate the annual aggregate benefit implied by household valuation as:

$$B = 52 \times \overline{WTP}_{week} \times N$$

²The Wang-type MBDC estimate implies an average WTP of about 22 CNY per household per week, as reported in Appendix Table C1. Multiplying by 52 weeks and dividing by 12 months gives an implied monthly WTP of approximately 95 CNY. Average monthly household income in our sample is about 12,200 CNY, as shown in the descriptive statistics in Table 3.

We use the mean WTP from the Wang-type MBDC model, which is approximately 22 CNY per household per week, as reported in Appendix Table C1. We set $N = 5$ million based on Xi'an's 2024 population of about 13 million and the average household size of 2.62 reported in the 7th national population census. This assumption provides a conservative estimate of aggregate household valuation since the project also benefits surrounding cities beyond Xi'an.

We compare the present value of these benefits with the construction cost of the project. Let C_0 denote the initial construction investment, and let H denote the project horizon. The net benefit at horizon H is defined as:

$$NB(H) = -C_0 + \sum_{t=1}^H \frac{B}{(1+r)^t}. \quad (7)$$

We set $C_0 = 51.6$ billion CNY³ and use a benchmark discount rate of 8 percent⁴. We also vary the discount rate from 3 to 10 percent as the sensitivity analysis. In equation (7), we do not separately include annual operation and maintenance costs, since a substantial share of these costs is expected to be covered through the official water supply price set by the government⁵.

Figure 3 plots the net benefits against the project horizon. Net benefits increase as discounted annual household valuations accumulate over time and turn positive after approximately 17 years. This break-even point suggests that the aggregate valuation implied by our WTP estimate can cover the construction cost within a plausible horizon for long-lived water infrastructure.

However, this analysis should be interpreted with caution. First, the aggregate benefit depends on the assumed beneficiary population. Using only Xi'an households provides a conservative estimate that is closer to the survey population, while using the broader Guanzhong Plain better reflects the geographic scope of the project but requires stronger external validity assumptions. Second, WTP should be interpreted as perceived household valuation of the project. It may not fully capture broader economic benefits generated outside the surveyed population, such as regional development, industrial expansion, or economy-wide productivity gains. Therefore, the true net social benefits may be larger than our baseline estimate.

³This number is from government annual budget reports and the construction company.

⁴This is the benchmark social discount rate established by the National Development and Reform Commission and the Ministry of Construction in 2006 for the economic evaluation of construction projects.

⁵The Shaanxi Development and Reform Commission set the trial water supply price from the water basins to local utilities at 1.23 CNY per cubic meter.

5 Robustness Checks and Additional Discussions

In this section, we present robustness checks and additional findings. First, we examine the sensitivity of our estimates by adopting alternative distributional assumptions in the estimation. Second, we assess the robustness to the probability assignment used in the Wang-type MBDC model. Third, we provide a brief discussion of the role of the oath script in our survey design.

5.1 Robustness Checks

Alternative parametric assumptions. Although misspecification of the parametric distribution in maximum likelihood estimation does not necessarily invalidate consistency under standard regularity conditions, we assess the robustness of our results by re-estimating all models under a normal distribution assumption. The corresponding estimates are reported in Appendix Tables [D1](#) to [D4](#). We find no substantive differences in the coefficient estimates or in the implied WTP distributions, indicating that our results are robust to alternative distributional assumptions.

Alternative probability assignments. We check the sensitivity of estimates to the probability assignment in the Wang-type model by considering two alternative probability sets of (0.9, 0.7, 0.5, 0.3, 0.1) and (0.95, 0.7, 0.5, 0.3, 0.05). Detailed estimates are in Appendix Tables [E1](#) to [E4](#), and the estimates are similar to the baseline results.

5.2 Effectiveness of Oath Scripts

As part of our survey design, respondents were required to sign a brief commitment statement before proceeding to the questionnaire. The inclusion of such oath scripts has been argued to reduce the gap between the true preferences of the respondents and their stated preferences by increasing the perceived moral and psychological cost of misreporting, and it is known that the direction and magnitude of this hypothetical bias vary with cultural norms, education, and broader social contexts ([Howard et al., 2017](#); [Vossler et al., 2025](#)).

To account for the oath script, we supplement the main sample with an additional group of 89 respondents who completed the MBDC questionnaires without being exposed to the oath script. We then re-estimate the Wang-type model parameters using maximum likelihood, including an indicator for oath script exposure as an additional covariate. The estimation results are reported

in Appendix Table F1. The corresponding WTP distribution is shown in Figure 2 (f). We find that the inclusion of an oath script is significantly related to a lower stated WTP in Appendix Table F1. The KS tests show that the distributions are significantly different at the 5% level⁶. This downward adjustment of hypothetical WTP is consistent with the growing literature documenting that oath scripts can substantially mitigate hypothetical bias and reduce overstatement in stated-preference surveys for public-interest goods (Penn and Hu, 2019; Schmidt and Bijmolt, 2020).

It is important to note that all cross-format comparisons reported earlier are based on the main survey design in which the oath script was implemented. Given the evidence in Appendix Table F1 that the oath script is significantly related to lower WTP, our primary results reflect a survey environment in which hypothetical bias is already mitigated. The persistence of strong agreement across elicitation formats under these conditions suggests that the consistency of welfare estimates documented above is unlikely to be driven by hypothetical bias. Instead, any remaining bias appears to affect the overall level of stated values rather than the relative performance of alternative valuation models.

6 Conclusion and Implications

This paper compares three widely used CV models: DBDC, MBDC with a Welsh-type specification (Welsh and Poe, 1998), and MBDC with a Wang-type specification (Wang and He, 2011). Using survey data from a large-scale inter-basin water diversion project in China, we estimate household WTP under maximum likelihood estimation and assess consistency across methods.

Across all three models, the determinants of WTP are broadly consistent. Income, gender, age, perceived water scarcity, water quality, and knowledge of water systems robustly affect valuation outcomes. Although the MBDC specifications reveal additional heterogeneity, particularly by education and water-related perceptions, the main conclusions regarding WTP determinants remain stable across elicitation formats.

Once differences in bid support are accounted for, the three approaches deliver closely aligned WTP distributions. In particular, the DBDC switching interval aligns with the MBDC uncertainty region, especially the Not sure category, suggesting that these response formats locate a similar

⁶The KS statistic is 1.40.

underlying valuation boundary. Consistent with the visual evidence, the Kolmogorov–Smirnov and Wald tests based on a common interval representation do not reject equality across models. To the extent that uncertainty responses are informative about respondents’ underlying WTP, this convergence indicates that alternative CVM formats can yield broadly comparable welfare estimates when bid support and other survey-design choices are carefully aligned.

These findings offer several practical implications for applied CVM studies. First, comparisons of WTP across elicitation formats should carefully account for bid range selection, since apparent differences may largely reflect differences in support rather than preferences. Second, distributional comparisons provide a more informative robustness assessment than comparisons based only on mean WTP. Third, richer elicitation formats do not automatically yield more precise estimates: models that jointly exploit multiple valuation cutoffs within a coherent likelihood framework can outperform two-step alternatives. Thus, when policy evaluation requires precise welfare measures with limited sample sizes, relatively simple designs with carefully calibrated bids may be preferable to more complex formats. These considerations are particularly important when CVM estimates are used in cost-benefit analysis.

Several limitations remain. First, the analysis focuses on one urban setting and one large-scale water infrastructure project, which may limit the external validity of the estimated WTP. Second, as in other stated-preference studies, the estimates may be subject to hypothetical bias; however, the oath script and robustness checks help mitigate this concern. Third, even though the supplementary sample improves comparability by harmonizing bid ranges, bid design remains an important determinant of valuation outcomes. Finally, the Wang-type MBDC model relies on subjective probability assignments for ordered responses; alternative mappings may generate different efficiency properties. Future research could examine these issues in other policy settings, experimentally vary information and bid structures, and integrate revealed-preference data with CVM estimates.

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Tables

Table 1: DBDC Questionnaires

Group	First bid amount	Second bid amount (high)	Second bid amount (low)	Abbreviation
1	2	4	1	2-4-1
2	4	8	2	4-8-2
3	8	15	4	8-15-4
4	15	30	8	15-30-8

Table 2: MBDC Questionnaires

Question: for each of bid amount below, choose your level of willingness to pay.					
	Definitely yes	Probably yes	Not sure	Probably no	Definitely no
Bids (CNY)					
0					
0.5					
1					
1.5					
2					
3					
4					
5					
6					
7					
8					
10					
15					
20					
27					
35					
42					
50					
65					
80					
100					
150					
200					
250					
300					

Table 3: Descriptive Statistics

Variables	DBDC	MBDC	Between group t-statistics	City census
Respondent characteristics				
Age	46.11 (15.02)	45.99 (15.01)	0.13	42.97
Female	0.53 (0.49)	0.52 (0.48)	0.33	0.51
Hold a college degree	0.32 (0.46)	0.34 (0.47)	-0.68	0.30
Household characteristics				
Family size	3.48 (1.34)	3.39 (1.39)	1.04	2.90
Family monthly income per capita	3,511.22 (2,671.01)	3,431.82 (2,425.35)	0.50	3,568.17
Family monthly water bill	35.19 (17.06)	36.81 (16.26)	-1.54	
Installation of water filters	0.27 (0.44)	0.32 (0.39)	-1.92	
Water scarcity				
Score of tap water pressure (1 to 5: bad to good)	3.47 (0.77)	3.49 (0.71)	-0.43	
Frequency of water outage in the past year	0.16 (0.37)	0.20 (0.39)	-1.65	
Longest outage duration in the past year (hour)	0.86 (1.17)	0.80 (1.30)	0.76	
Water quality				
Score of tap water turbidity (1 to 5: bad to good)	3.22 (0.90)	3.10 (0.83)	2.21	
Score of tap water impurity (1 to 5: bad to good)	2.23 (0.91)	2.50 (0.82)	-4.97	
Score of overall tap water quality (1 to 10: bad to good)	3.25 (1.35)	3.32 (1.37)	-0.81	
Knowledge of water sources				
Number of correctly listed tap water sources in the city (0 to 4)	0.81 (0.82)	0.75 (0.80)	1.17	
Know the tap water sources at own home (binary)	0.21 (0.41)	0.25 (0.39)	-1.59	
Concern water pollution (binary)	0.77 (0.42)	0.71 (0.40)	2.32	
Know the pricing schedule of tap water (binary)	0.48 (0.50)	0.42 (0.52)	1.85	
Observations	610	422		

Note: This table reports descriptive statistics for respondents assigned to the DBDC and MBDC elicitation formats. Detailed categorical statistics are reported in Appendix Tables [A1](#) to [A3](#).

Table 4: DBDC Estimates

	Coef. (1)	Std. err. (2)
Respondent's characteristics		
Age	-0.007	(0.004)
Female	-0.349	(0.127)
College degree	0.147	(0.162)
Household characteristics		
Monthly income per capita (log)	0.014	(0.004)
Monthly water bill	0.011	(0.004)
Installation of water filters	0.791	(0.126)
Water usage		
Water scarcity index	0.391	(0.088)
Water quality index	-0.271	(0.074)
Water knowledge index	0.166	(0.046)
Other parameters		
Intercept	2.731	(0.531)
Std. err. ε_i	0.682	(0.042)
Observation	610	
Log likelihood	-683.994	

Note: This table reports DBDC estimates of households' WTP determinants. Coefficients are obtained from equation (1). Standard errors are in parentheses.

Table 5: MBDC Estimates using Welsh-type

	Definitely yes (1)	Probably yes (2)	Not sure (3)	Probably no (4)	Definitely no (5)
Respondent's characteristics					
Age	-0.013 (0.004)	-0.013 (0.004)	-0.016 (0.004)	-0.012 (0.004)	-0.015 (0.004)
Female	-0.161 (0.042)	-0.159 (0.038)	-0.158 (0.038)	-0.162 (0.040)	-0.163 (0.039)
College degree	0.306 (0.101)	0.314 (0.104)	0.331 (0.110)	0.355 (0.102)	0.342 (0.110)
Household characteristics					
Monthly income per capita (log)	0.019 (0.006)	0.021 (0.006)	0.022 (0.010)	0.021 (0.006)	0.021 (0.010)
Monthly water bill	0.002 (0.003)	0.001 (0.003)	0.000 (0.003)	0.000 (0.003)	0.001 (0.003)
Installation of water filters	0.057 (0.117)	0.077 (0.118)	0.057 (0.120)	0.069 (0.119)	0.055 (0.120)
Water usage					
Water scarcity index	0.162 (0.076)	0.153 (0.076)	0.163 (0.076)	0.152 (0.076)	0.160 (0.076)
Water quality index	-0.421 (0.078)	-0.538 (0.083)	-0.462 (0.079)	-0.496 (0.080)	-0.503 (0.079)
Water knowledge index	0.106 (0.021)	0.137 (0.025)	0.141 (0.020)	0.118 (0.017)	0.121 (0.025)
Other parameters					
Intercept	2.123 (0.452)	2.666 (0.592)	3.036 (0.710)	3.464 (0.721)	3.693 (0.804)
Std. err. ε_i	0.634 (0.201)	0.725 (0.203)	0.935 (0.282)	1.035 (0.321)	1.134 (0.325)
Observation	422				
Log likelihood	-1310.225	-1317.464	-1361.637	-1305.585	-1310.255

Note: This table reports Welsh-type MBDC estimates of households' WTP determinants. Coefficients are obtained from equation (2). Standard errors are in parentheses.

Table 6: MBDC Estimates using Wang-type Second Stage

	Coef. (1)	Std. err. (2)
Respondent's characteristics		
Age	-0.025	(0.005)
Female	-0.234	(0.067)
College degree	0.226	(0.061)
Household characteristics		
Monthly income per capita (log)	0.028	(0.013)
Monthly water bill	0.004	(0.004)
Installation of water filters	0.026	(0.161)
Water usage		
Water scarcity index	0.104	(0.028)
Water quality index	-0.423	(0.102)
Water knowledge index	0.208	(0.052)
Other parameter		
Intercept	2.841	(0.725)
Observation	422	
R-squared	0.828	

Note: This table reports Wang-type MBDC second-stage estimates of households' WTP determinants. Coefficients are obtained from equation (4). Standard errors are in parentheses.

Table 7: KS statistics

Welsh-type:	Definitely yes	Probably yes	Not sure	Probably no	Definitely no
Original MBDC Group					
DBDC	1.78	1.29	1.45	1.72	1.84
Wang-type	1.98	1.28	1.08	1.43	1.69
KS test between DBDC and Wang-type: 1.66					
Capped MBDC Group					
DBDC	1.74	1.47	1.12	1.42	1.71
Wang-type	1.81	1.52	1.15	1.44	1.67
KS test between DBDC and Wang-type: 1.20					

Note: This table reports KS statistics comparing the WTP distributions implied by different models. $p < 0.1$: KS statistics > 1.22 ; $p < 0.05$: KS statistics > 1.36 ; $p < 0.01$: KS statistics > 1.62 , based on bootstrap.

Table 8: Wald statistics

Welsh-type:	Definitely yes	Probably yes	Not sure	Probably no	Definitely no
Original MBDC Group					
DBDC	24.73	21.14	25.92	31.45	34.28
Wang-type	33.67	23.81	15.62	24.94	30.16
Wald test between DBDC and Wang-type: 16.21					
Capped MBDC Group					
DBDC	32.48	24.05	17.44	23.76	31.08
Wang-type	34.12	25.11	18.73	24.69	29.84
Wald test between DBDC and Wang-type: 14.36					

Note: This table reports Wald statistics comparing the three specifications. $p < 0.1$: Wald statistics > 21.06 ; $p < 0.05$: Wald statistics > 23.68 ; $p < 0.01$: Wald statistics > 29.14 , based on bootstrap.

Table 9: Comparison with Existing Water Infrastructure Valuation Studies

Study	Project	Format	Estimated valuation
Moreno-Sanchez et al. (2012)	Watershed protection, Colombia	CVM PC	0.28% of monthly household income
Tanellari et al. (2015)	Drinking water quality and infrastructure, U.S.	DCE	0.37% of monthly household income
Chatterjee et al. (2017)	Tap water quality, U.S.	CVM PC	0.11% of monthly household income
Guerrini et al. (2018)	Water sources, networks, and wastewater treatment, Italy	CVM PC	0.01% of average monthly income
Abu Hatab et al. (2025)	Irrigation water supply, Egypt	CVM DBDC	0.58% of monthly income per feddan
Sone et al. (2025)	Upstream NbS for water security, Brazil	DCE	almost 1% of monthly disposable household income
Moon and Choi (2026)	Industrial desalination, South Korea	CVM DBDC	0.03% of monthly household income
Sarkar and Lupi (2026)	Water/sewer infrastructure and affordability, U.S.	DCE	0.29% for infrastructure; 0.12% for affordability
Lambert et al. (2026)	Municipal water reuse, U.S.	CVM PC	0.19% of monthly household income

Note: PC is for payment card. DCE is for discrete choice experiment.

Figures

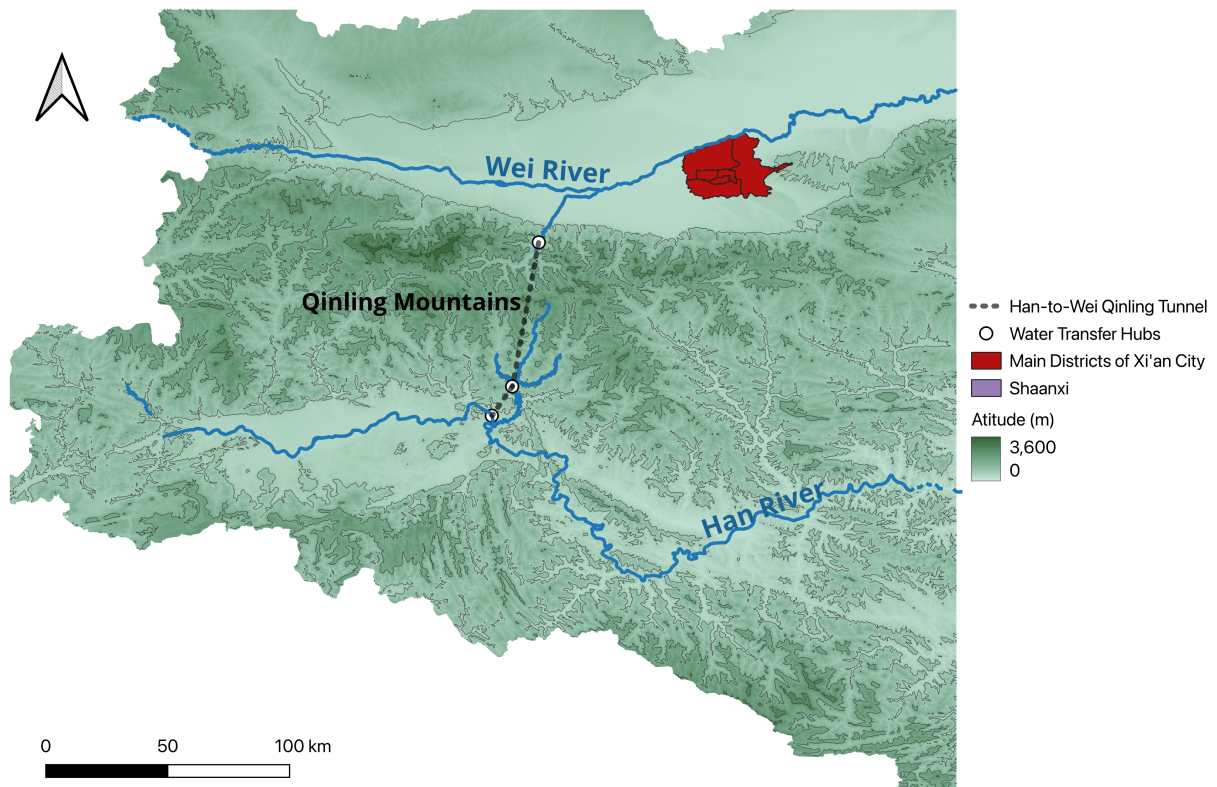


Figure 1: The Map of Han-to-Wei River Water Diversion Project

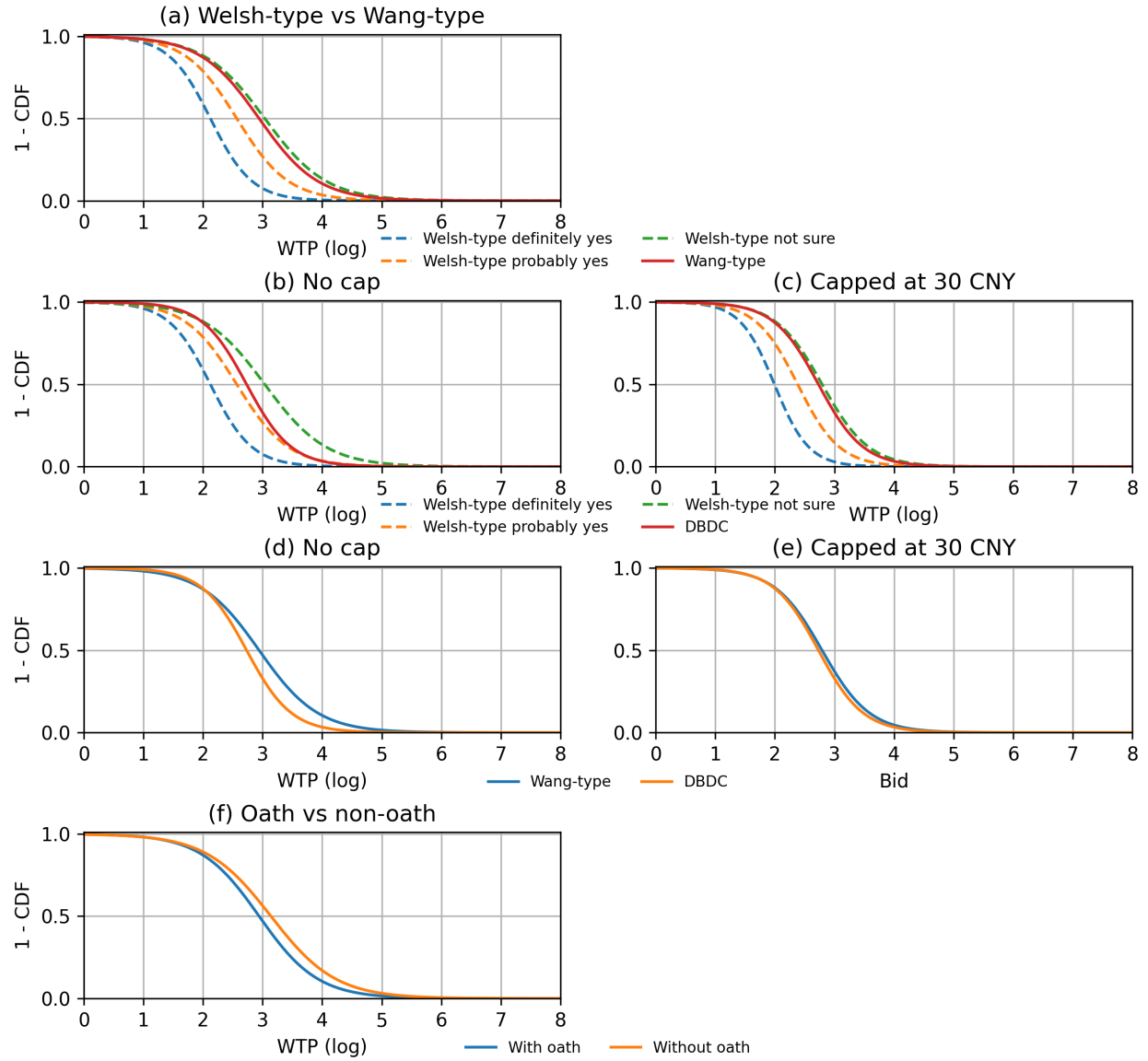


Figure 2: The Distribution of Estimated WTP (log)

Note: This figure shows the implied WTP (log) distribution (survival function) of each valuation model and scenario. (a) to (e) are pairwise comparison. (a), (b), and (d) use the MBDC sample with bids up to 300 CNY. (c) and (e) use the additional MBDC sample capped at 30 CNY. (f) reports the distribution with or without oath script using Wang-type model.

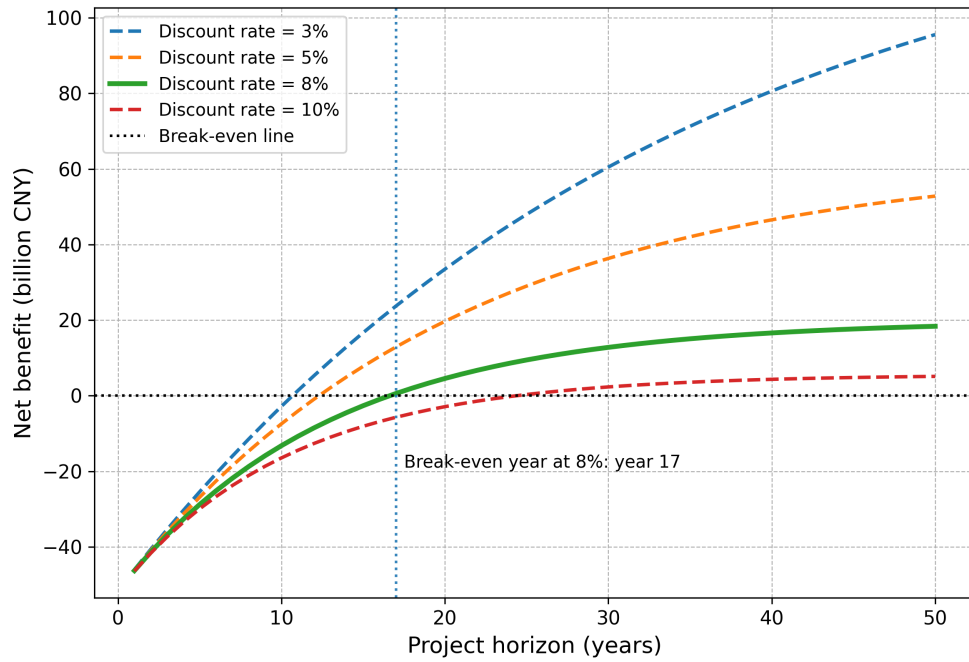


Figure 3: Net Benefit of the Project

Note: This figure reports the simulated cost-benefit analysis. Net benefits are calculated using mean weekly WTP from the Wang-type MBDC model, 5 million beneficiary households, a construction cost of 51.6 billion CNY, and alternative percent discount rates with 8 percent as the baseline.

Appendix

A Descriptive Statistics

Table A1: Categorical Descriptive Statistics (1 of 3)

Variables	DBDC	MBDC
Respondent's characteristics		
Age		
0-24 years old	0.071	0.062
25-39 years old	0.268	0.271
40-65 years old	0.507	0.492
Above 65 years old	0.154	0.175
Household characteristics		
Family monthly income per capita		
0-500	0.040	0.039
500-1,000	0.107	0.135
1,000-2,000	0.296	0.271
2,000-4,000	0.398	0.372
4,000-6,000	0.107	0.124
Above 6,000	0.073	0.059

Note: This table reports categorical statistics for demographic variables.

Table A2: Categorical Descriptive Statistics (2 of 3)

Variables	DBDC	MBDC
Water scarcity		
Score of tap water pressure (1 to 5: bad to good)		
Score 1	0.010	0.010
Score 2	0.239	0.229
Score 3	0.288	0.292
Score 4	0.196	0.201
Score 5	0.267	0.268
Frequency of water outage in the past year		
Freq. 0	0.533	0.524
Freq. 1-2	0.393	0.368
Freq. above 2	0.074	0.108
Longest outage duration in the past year (hour)		
Less than 1	0.528	0.538
1-2	0.381	0.391
Above 3	0.091	0.071

Note: This table reports categorical statistics for water scarcity.

Table A3: Categorical Descriptive Statistics (3 of 3)

Variables	DBDC	MBDC
Water quality		
Score of tap water turbidity (1 to 5: bad to good)		
Score 1	0.040	0.050
Score 2	0.180	0.220
Score 3	0.350	0.360
Score 4	0.380	0.320
Score 5	0.050	0.050
Score of tap water impurity (1 to 5: bad to good)		
Score 1	0.200	0.120
Score 2	0.450	0.400
Score 3	0.290	0.360
Score 4	0.040	0.100
Score 5	0.020	0.020
Score of overall tap water quality (1 to 10: bad to good)		
Score 1-2	0.445	0.420
Score 3-4	0.325	0.340
Score 5-6	0.160	0.170
Score 7-8	0.050	0.050
Score 9-10	0.020	0.020

Note: This table reports categorical statistics for water quality.

Table A4: Distribution of DBDC Responses

Bids	YY (%)	YN (%)	NY (%)	NN (%)
2-4-1	80.5	10.9	2.7	5.9
4-8-2	61.2	19.5	8.6	10.7
8-15-4	50.9	22.9	9.9	16.3
15-30-8	27.1	28.7	20.8	23.4

Note: YY, YN, NY, and NN denote response options in the DBDC format, where Y and N indicate yes and no responses to sequential bid offers.

Table A5: Distribution of MBDC Responses

Bids (CNY)	Definitely yes (%)	Probably yes (%)	Not sure (%)	Probably not (%)	Definitely not (%)
0	100.0	0.0	0.0	0.0	0.0
0.5	87.4	7.8	0.0	0.2	4.5
1.0	82.5	10.7	2.1	0.0	4.7
1.5	77.3	11.4	4.5	1.7	5.2
2.0	72.7	14.0	4.5	2.4	6.4
3.0	62.8	16.6	8.8	3.8	8.1
4.0	57.3	16.4	9.5	7.3	9.5
5.0	54.0	14.5	10.9	7.8	12.8
6.0	43.4	14.5	14.0	8.1	20.1
7.0	42.2	12.6	11.6	9.7	23.9
8.0	40.8	10.7	11.4	9.0	28.2
10	37.4	11.6	7.8	9.0	34.1
15	18.7	20.4	10.9	9.0	41.0
20	15.9	12.6	12.8	11.4	47.4
27	8.8	12.1	10.2	13.7	55.2
35	6.6	9.2	11.4	10.4	62.3
42	5.0	6.2	8.5	12.1	68.2
50	4.7	5.0	6.6	8.8	74.9
65	2.1	4.0	3.8	8.8	81.3
80	1.9	1.9	4.3	6.2	85.8
100	1.2	2.1	2.6	4.0	90.0
150	0.7	1.4	1.7	2.6	93.6
200	0.7	1.2	0.5	2.1	95.5
250	0.7	0.9	0.5	0.5	97.4
300	0.7	0.9	0.2	0.0	98.1

Note: This table reports the percentage distribution of respondents across five uncertainty categories at each bid amount under MBDC format.

B Principle Component Analysis

Table B1: PCA for Water Scarcity Index

Variable	DBDC			MDBC		
	PC1	PC2	PC3	PC1	PC2	PC3
Water pressure	-0.277	-0.939	0.202	-0.305	-0.826	0.193
Outage frequency	0.698	0.052	-0.714	0.638	0.105	-0.628
Outage duration	0.660	0.339	0.671	0.625	0.407	0.618
Eigenvalue	1.577	0.976	0.447	1.617	0.925	0.459
Proportion explained	0.526	0.325	0.149	0.539	0.308	0.153
Cumulative explained	0.526	0.851	1.000	0.539	0.847	1.000

Note: This table reports PCA results used to construct the water scarcity index. The first principal component is used as the water scarcity index in the main analysis.

Table B2: PCA for Water Quality Index

Variable	DBDC			MDBC		
	PC1	PC2	PC3	PC1	PC2	PC3
Water turbidity	0.608	-0.294	-0.736	0.539	-0.286	-0.762
Water impurity	0.594	-0.445	0.669	0.462	-0.436	0.681
Water quality score	0.525	0.845	0.096	0.582	0.804	0.105
Eigenvalue	1.865	0.673	0.461	2.145	0.546	0.309
Proportion explained	0.621	0.224	0.153	0.715	0.182	0.103
Cumulative explained	0.621	0.846	1.000	0.715	0.897	1.000

Note: This table reports PCA results used to construct the water quality index. The first principal component is used as the water quality index in the main analysis.

Table B3: PCA for Water Knowledge

Variable	DBDC				MDBC			
	PC1	PC2	PC3	PC4	PC1	PC2	PC3	PC4
Number of known water sources	0.629	-0.261	-0.137	-0.718	0.682	-0.243	-0.167	-0.709
Knows water source	0.601	-0.405	-0.059	0.685	0.617	-0.417	-0.082	0.638
Concern water pollution	0.315	0.746	-0.574	0.114	0.325	0.738	-0.561	0.138
Knows water price	0.377	0.458	0.804	0.010	0.338	0.462	0.828	0.017
Eigenvalue	1.588	0.977	0.877	0.556	1.312	1.180	0.952	0.556
Proportion explained	0.397	0.244	0.219	0.139	0.328	0.295	0.238	0.139
Cumulative explained	0.397	0.641	0.861	1.000	0.328	0.623	0.861	1.000

Note: This table reports PCA results used to construct the water knowledge index. The first principal component is used as the water knowledge index in the main analysis.

C Wang-type First Stage

Table C1: MBDC Estimates with Wang-type First Stage

	Mean (1)	Std. dev. (2)	Min. (3)	Max. (4)
Estimated WTP (μ_i)	22.54	20.41	0.06	233.10
Estimated WTP std. dev. (σ_i)	9.84	14.96	0.02	177.31
Observation	422			

Note: This table reports first-stage estimates of each individual's WTP under the Wang-type MBDC specification. The individual-level mean and standard deviation of WTP are obtained from equation (3). We show the WTP instead of presenting the log form.

D Normal Distribution

Table D1: DBDC Estimates

	Coef. (1)	Std. err. (2)
Respondent's characteristics		
Age	-0.026	(0.007)
Female	-0.238	(0.046)
College degree	0.219	(0.069)
Household characteristics		
Monthly income per capita (log)	0.027	(0.013)
Monthly water bill	0.003	(0.005)
Installation of water filters	0.034	(0.167)
Water usage		
Water scarcity index	0.111	(0.036)
Water quality index	-0.438	(0.121)
Water knowledge index	0.197	(0.064)
Other parameter		
Intercept	2.918	(0.847)
Observation	610	
Log likelihood	-682.598	

Note: This table reports DBDC estimates of households' WTP determinants assuming a normal distribution. Coefficients are obtained from equation (1). Standard errors are in parentheses.

Table D2: MBDC Estimates using Welsh-type

	Definitely yes (1)	Probably yes (2)	Not sure (3)	Probably no (4)	Definitely no (5)
Respondent's characteristics					
Age	-0.014 (0.005)	-0.012 (0.004)	-0.017 (0.005)	-0.013 (0.004)	-0.016 (0.005)
Female	-0.154 (0.045)	-0.167 (0.041)	-0.160 (0.042)	-0.158 (0.043)	-0.169 (0.041)
College degree	0.298 (0.098)	0.322 (0.107)	0.339 (0.113)	0.347 (0.104)	0.336 (0.111)
Household characteristics					
Monthly income per capita (log)	0.020 (0.007)	0.022 (0.006)	0.023 (0.011)	0.020 (0.007)	0.022 (0.010)
Monthly water bill	0.003 (0.004)	0.002 (0.003)	0.001 (0.003)	0.001 (0.003)	0.002 (0.004)
Installation of water filters	0.062 (0.121)	0.071 (0.122)	0.059 (0.124)	0.073 (0.120)	0.058 (0.123)
Water usage					
Water scarcity index	0.168 (0.079)	0.149 (0.078)	0.159 (0.077)	0.155 (0.078)	0.164 (0.079)
Water quality index	-0.432 (0.081)	-0.524 (0.086)	-0.471 (0.082)	-0.487 (0.083)	-0.495 (0.082)
Water knowledge index	0.112 (0.023)	0.133 (0.027)	0.145 (0.022)	0.119 (0.019)	0.126 (0.026)
Other parameters					
Intercept	2.201 (0.471)	2.598 (0.613)	3.118 (0.732)	3.402 (0.744)	3.742 (0.821)
Std. err. ε_i	0.648 (0.209)	0.739 (0.214)	0.921 (0.295)	1.048 (0.337)	1.152 (0.338)
Observation	422				
Log likelihood	-1315.284	-1314.267	-1368.249	-1312.214	-1311.284

Note: This table reports Welsh-type MBDC estimates of households' WTP determinants assuming normal distribution. Coefficients are obtained from equation (2). Standard errors are in parentheses.

Table D3: MBDC Estimates using Wang-type First Stage

	Mean (1)	Std. dev. (2)	Min. (3)	Max. (4)
Estimated WTP (μ_i)	22.50	30.46	0.07	233.14
Estimated WTP std. dev. (σ_i)	9.88	14.90	0.03	177.28
Observation	422			

Note: This table reports first-stage estimates of each individual's WTP under the Wang-type MBDC specification with the normal distribution. The individual-level mean and standard deviation of WTP are obtained from equation (3). We show the WTP instead of presenting the log form.

Table D4: MBDC Estimates using Wang-type Second Stage

	Coef. (1)	Std. err. (2)
Respondent's characteristics		
Age	-0.024	(0.006)
Female	-0.221	(0.072)
College degree	0.233	(0.067)
Household characteristics		
Monthly income per capita (log)	0.027	(0.014)
Monthly water bill	0.003	(0.005)
Installation of water filters	0.031	(0.168)
Water usage		
Water scarcity index	0.109	(0.032)
Water quality index	-0.417	(0.118)
Water knowledge index	0.203	(0.064)
Other parameter		
Intercept	2.764	(0.812)
Observation	422	
R-squared	0.829	

Note: This table reports Wang-type MBDC second-stage estimates of households' WTP determinants assuming normal distribution. Coefficients are obtained from equation (4). Standard errors are in parentheses.

E Alternative Probability Assignments

Table E1: MBDC Estimates using Wang-type First Stage

	Mean (1)	Std. dev. (2)	Min. (3)	Max. (4)
Estimated WTP (μ_i)	22.54	30.50	0.06	233.18
Estimated WTP std. dev. (σ_i)	9.93	14.94	0.04	177.33
Observation	422			

Note: This table reports first-stage estimates of each individual's WTP under the Wang-type MBDC specification with probabilities of 0.9, 0.7, 0.5, 0.3, and 0.1. The individual-level mean and standard deviation of WTP are obtained from equation (3). We show the WTP instead of presenting the log form.

Table E2: MBDC Estimates using Wang-type Second Stage

	Coef. (1)	Std. err. (2)
Respondent's characteristics		
Age	-0.026	(0.007)
Female	-0.242	(0.074)
College degree	0.219	(0.064)
Household characteristics		
Monthly income per capita (log)	0.029	(0.014)
Monthly water bill	0.003	(0.005)
Installation of water filters	0.033	(0.173)
Water usage		
Water scarcity index	0.111	(0.033)
Water quality index	-0.439	(0.121)
Water knowledge index	0.214	(0.068)
Other parameter		
Intercept	2.903	(0.847)
Observation	422	
R-squared	0.818	

Note: This table reports Wang-type MBDC second-stage estimates of households' WTP determinants with probabilities of 0.9, 0.7, 0.5, 0.3, and 0.1. Coefficients are obtained from equation (4). Standard errors are in parentheses.

Table E3: MBDC Estimates using Wang-type First Stage

	Mean (1)	Std. dev. (2)	Min. (3)	Max. (4)
Estimated WTP (μ_i)	22.48	30.43	0.08	233.11
Estimated WTP std. dev. (σ_i)	9.86	14.93	0.05	177.26
Observation	422			

Note: This table reports first-stage estimates of each individual's WTP under the Wang-type MBDC specification with probabilities of 0.95, 0.7, 0.5, 0.3, and 0.05. The individual-level mean and standard deviation of WTP are obtained from equation (3). We show the WTP instead of presenting the log form.

Table E4: MBDC Estimates using Wang-type Second Stage

	Coef. (1)	Std. err. (2)
Respondent's characteristics		
Age	-0.027	(0.008)
Female	-0.229	(0.071)
College degree	0.241	(0.069)
Household characteristics		
Monthly income per capita (log)	0.026	(0.013)
Monthly water bill	0.003	(0.004)
Installation of water filters	0.029	(0.169)
Water usage		
Water scarcity index	0.113	(0.035)
Water quality index	-0.411	(0.115)
Water knowledge index	0.198	(0.063)
Other parameter		
Intercept	2.912	(0.831)
Observation	422	
R-squared	0.804	

Note: This table reports Wang-type MBDC second-stage estimates of households' WTP determinants with probabilities of 0.95, 0.7, 0.5, 0.3, and 0.05. Coefficients are obtained from equation (4). Standard errors are in parentheses.

F Oath Script

Table F1: MBDC Estimates and Oath Scripts

	Coef. (1)	Std. err. (2)
Oath scripts		
Oath scripts	-0.128	(0.028)
Respondent's characteristics		
Age	-0.024	(0.006)
Female	-0.221	(0.044)
College degree	0.231	(0.073)
Household characteristics		
Monthly income per capita (log)	0.029	(0.014)
Monthly water bill	0.005	(0.005)
Installation of water filters	0.031	(0.158)
Water usage		
Water scarcity index	0.109	(0.034)
Water quality index	-0.417	(0.119)
Water knowledge index	0.203	(0.067)
Other parameter		
Intercept	2.764	(0.812)
Observation	511	
R-squared	0.863	

Note: This table reports Wang-type MBDC second-stage estimates of households' WTP determinants with the inclusion of oath scripts. Coefficients are obtained from equation (4). Standard errors are in parentheses.

G Statistical Efficiency of Estimates

As the final component of the comparison, we evaluate the relative statistical efficiency of the three estimation models using the trace (A-optimality) and the determinant (D-optimality) criteria (Dette, 1997; Jones et al., 2021). Because the parameter vector β exists in all specifications, the comparison is based on the estimated covariance matrix of $\hat{\beta}$. Specifically, we consider the following two statistics.

$$\text{tr}(\hat{\Sigma}_\beta) \quad \text{and} \quad \det(\hat{\Sigma}_\beta)^{1/k_\beta}$$

where $\text{tr}(\cdot)$ denotes the trace, $\det(\cdot)$ denotes the determinant, and k_β is the dimension of the parameter vector β . The matrix $\hat{\Sigma}_\beta$ is defined as:

$$\hat{\Sigma}_\beta = n \cdot \widehat{\text{Var}}(\hat{\beta})$$

which scales the estimated covariance matrix by the sample size for consistent comparisons across estimation models. Smaller values of either criterion indicate greater statistical efficiency.

The covariance matrix $\hat{\Sigma}_\beta$ is derived from the Fisher information matrix according to the DBDC and Welsh MBDC specifications by a one-step maximum likelihood estimation. In contrast, under the Wang-type MBDC approach, which follows a two-step estimation procedure, $\hat{\Sigma}_\beta$ is obtained from the corresponding Godambe information and takes a sandwich form. To ensure comparability between models and to accommodate potential deviations from parametric assumptions, we compute $\hat{\Sigma}_\beta$ using a bootstrap procedure for all three specifications.

We show the comparison of statistical efficiency in Appendix Table G1. The Welsh-type MBDC specification is the most efficient because it uses multiple valuation cutoffs per respondent and estimates all parameters jointly in a single-step maximum likelihood framework. The DBDC approach is less efficient than the Welsh-type MBDC because each respondent provides only two valuation points, but it remains relatively efficient due to its single-step estimation and direct use of the Fisher information matrix. In contrast, the Wang-type MBDC approach exhibits lower efficiency because it relies on a two-step procedure with generated regressors, and inference is based on the Godambe and sandwich form rather than the complete Fisher information matrix, resulting in larger estimated variances for β .

Table G1: Relative Statistical Efficiency Compared to DBDC

	$\text{tr}(\widehat{\Sigma}_\beta)$	$\det(\widehat{\Sigma}_\beta)^{1/k_\beta}$
Welsh-type MBDC		
Definitely yes	0.62	0.70
Probably yes	0.65	0.72
Not sure	0.64	0.77
Probably no	0.60	0.71
Definitely no	0.67	0.79
Wang-type MBDC	1.21	1.27
Normalized DBDC	1.00	1.00

Note: This table reports relative statistical efficiency measures for the Welsh-type and Wang-type MBDC estimates, normalized to the DBDC benchmark. The trace and determinant of the estimated covariance matrix for the DBDC specification are normalized to one. Lower values indicate higher statistical efficiency relative to the DBDC baseline.